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Digital biomarkers for precision diagnosis and monitoring in Parkinson's disease

Yue-meng Sun¹, Zhi-yun Wang¹, Yuan-yuan Liang¹, Chen-wei Hao¹ & Chang-he Shi^{1,2,3,4} ✉

Parkinson's disease (PD) is a multifactorial neurodegenerative disorder with high prevalence among the elderly, primarily manifested by progressive decline in motor function. The aging global demographic and increased life expectancy have led to a rapid surge in PD cases, imposing a significant societal burden. PD along with other neurodegenerative diseases has garnered increasing attention from the scientific community. In PD, motor symptoms are recognized when approximately 60% of dopaminergic neurons have been damaged. The irreversible feature of PD and benefits of early intervention underscore the importance of disease onset prediction and prompt diagnosis. The advent of digital health technology in recent years has elevated the role of digital biomarkers in precisely and sensitively detecting early PD clinical symptoms, evaluating treatment effectiveness, and guiding clinical medication, focusing especially on motor function, responsiveness and sleep quality assessments. This review examines prevalent digital biomarkers for PD and highlights the latest advancements.

Parkinson's disease (PD) stands as the most prevalent movement disorder and the second most common degenerative ailment of the central nervous system, following Alzheimer's disease. The prevalence of PD is around 1% after the age of 60, escalating to 3% after 80¹. Its neuropathology primarily involves the formation of Lewy bodies (LB) in the substantia nigra (SN) pars compacta (SNpc) and the loss of dopaminergic (DAergic) neurons². While most PD patients exhibit bradykinesia, resting tremor and muscle rigidity, they also often endure a complex non-motor symptom (NMS) syndrome, encompassing anxiety, depression, sleep disorders and sensory symptoms³. The International Parkinson and Movement Disorders Society (MDS) categorizes early PD into three stages based on lesion, clinical signs or symptoms and diagnostic criteria: preclinical PD, prodromal PD and clinical PD⁴. Diagnosis predominantly relies on medical history and physical examination. The Unified Parkinson's Disease Rating Scale (UPDRS)⁵, developed in 1984 and later revised as MDS-UPDRS in 2008, offers a standardized method of PD assessment. Present treatment modalities solely provide symptomatic relief, as PD remains incurable, contributing to its high prevalence, significant disability and chronic nature. Throughout the progression of this chronic, lifelong disease, an extensive accumulation of medical data occurs, requiring significant resources for effective collection and analysis. Advances in mobile digital technologies have led to the development of innovative digital biomarkers, characterized by multi-dimensional measurements across diverse hardware and software layers.

These biomarkers facilitate sustained monitoring and documentation that extend beyond the traditional physical boundaries of clinical environments⁶. Amidst the COVID-19 pandemic, hospitals have integrated digital medical technologies and platforms, which facilitate disease progression monitoring and provide healthcare professionals with pertinent information for efficacy assessment. This narrative review aims to provide a comprehensive examination of digital biomarkers in PD. We conducted a systematic search across major academic databases and platforms, utilizing specific keywords associated with digital biomarkers in conjunction with PD. These included metrics for motor function, responsiveness, sleep and other relevant disturbances. Our selection criteria were focused on studies that utilize portable or home-based devices to measure these specific digital biomarkers of PD.

Digital biomarkers

Definition of digital biomarkers

In 2001, the U.S. National Institutes of Health formally defined a biomarker as "a characteristic that is objectively quantified and assessed as an indicator of normal biological functions, pathological states, or pharmacological responses to therapeutic interventions"⁷. Various molecules, including nucleic acids, proteins and metabolic derivatives, may serve as potential biomarkers⁸. Detecting PD during its initial stages is difficult and the condition is irreversible. With the rapid advancement of digital technology, a novel class of biomarkers, known as digital biomarkers, have emerged.

¹Department of Neurology, The First Affiliated Hospital of Zhengzhou University, Zhengzhou University, Zhengzhou, 450000 Henan, China. ²Henan Key Laboratory of Cerebrovascular Diseases, The First Affiliated Hospital of Zhengzhou University, Zhengzhou University, Zhengzhou, 450000 Henan, China. ³Institute of Neuroscience, Zhengzhou University, Zhengzhou, 450000 Henan, China. ⁴NHC Key Laboratory of Prevention and treatment of Cerebrovascular Diseases, The First Affiliated Hospital of Zhengzhou University, Zhengzhou University, Zhengzhou, 450000 Henan, China. ✉e-mail: shichanghe@gmail.com

“Digital” denotes the methodology of employing sensors and computational tools for data acquisition, which typically encompasses various hardware and software layers⁹. These digital biomarkers possess unique characteristics, such as the capability for longitudinal, continuous measurement and the generation of extensive datasets, sharing similar clinical objectives with traditional biomarkers but offering significant difference (Table 1). Digital biomarkers encompass objective, quantifiable physiological and behavioral measurements acquired through portable, wearable, implantable or ingestible digital devices¹⁰. These measurements are commonly performed using interconnected home products beyond the physical boundaries of clinical environments¹¹.

Measurement and collection of digital biomarkers

The measurement of digital biomarkers is inseparable from the development of digital health technology, catalyzing transformations in the medical domain¹². The measurement of digital biomarkers is characterized by its non-invasive nature, convenience, suitability for home use, and user-friendliness. Digital health devices employed for measuring digital biomarkers can be categorized into three main types. Firstly, wearable devices, such as wrist-worn accelerometers, portable sensors and biometric skin patches. Secondly, smartphones or applications are utilized for detecting voice/speech metrics, cognitive evaluations and typing behavior etc. Thirdly, non-visual technologies, which encompass passive measurement without requiring the subject to wear sensors, are implemented in sensor-equipped environments¹³. These devices can be worn on various parts of the human body or even within environments equipped with sensors (Fig. 1). Digital biomarker collection can be proactive, where individuals manually input data or perform designated tasks using digital devices. Alternatively, it can be passive, requiring no intervention from the device wearer and gathering data without disrupting the subject’s routine activities¹⁴, thereby enabling the acquisition of more continuous data.

Clinical digital biomarkers in Parkinson’s disease

Recently, there has been widespread research on digital biomarkers measured using smartphones and wearable devices in PD (Table 2). Furthermore, it can be utilized throughout various stages of PD (Fig. 2). This review provides an overview of the measurement of digital biomarkers for physiological function changes occurring in patients with PD.

Digital biomarkers in Parkinson’s disease
Serial reaction time

At the neural level, sequential motor learning is broadly distributed and forms the foundation for a series of motor, cognitive and social skills. It involves the capacity to execute a predefined sequence of movements with speed and precision effortlessly¹⁵. Serial reaction time (SRT) originally developed by Nissen and Bullemer has been utilized to investigate procedural memory learning in both clinical and non-clinical settings, commonly referred to as procedural learning, which is associated with the dorsolateral prefrontal cortex¹⁶. Participants are seated before a computer and prompted to react to recurring visual stimuli presented on the monitor by selecting one of several buttons on a response interface. This can distinguish PD patients

in the early stages of the disease and is influenced by dopamine medication therapy and disease progression¹⁷. The SRT task is categorized into single and dual tasks. Stephan et al. utilized a variant of the task, finding that sequence learning in PD patients was less effective and correlated with their disease stage¹⁸. SRT encompasses both implicit and explicit forms of sequence learning. Wilkinson et al. showed that, with extensive practice, PD patients could preserve diminished sequencing abilities¹⁹. To eliminate the influence of finger movements on SRT outcomes, Westwater et al. had participants verbally respond to stimuli²⁰. The SRT task, a prevalent tool for investigating sequence learning in PD, has compiled a significant body of data.

Sleep disturbances

Sleep disturbances rank among the most prevalent non-motor symptoms of PD, with an estimated 40–98% of PD patients globally experiencing such issues²¹, and frequently manifest well before the clinical diagnosis of PD. Laboratory-based video polysomnography (vPSG) remains the gold standard for evaluating sleep physiology in both normal and pathological conditions. Nevertheless, the substantial costs and inconvenience associated with vPSG have necessitated the development of alternative systems suitable for home use²². Technologies associated with wearable and portable devices now provide an alternative platform for monitoring sleep at home, leading to the development of head-mounted, wrist-worn and home-based devices²³.

González et al. utilized the Dreem2 EEG headband to measure overall sleep, efficiency, deep sleep, and rapid eye movement sleep in PD patients, achieving a specificity ranging from 78% to 96% due to its high accuracy in sleep classification²⁴. Additionally, Yang et al. demonstrated the feasibility of assessing PD in home settings by employing artificial intelligence modeling to analyze nocturnal respiratory signals collected from respiratory belts and radiofrequency-based respiratory monitoring devices²⁵. Actigraphy, employing a watch-like device with an integrated accelerometer, effectively monitors vital parameters²⁶. Ko et al. extracted the RR interval from the photoplethysmogram (PPG) sensor of a smartwatch, substantiating the utility of actigraphy in analyzing sleep stages among PD patients²⁷. Oñate-López et al. have developed smart bed sensors aimed at detecting sleep disturbances and enhancing sleep quality and user experience, although the current efficacy of these sensors remains to be established²⁸.

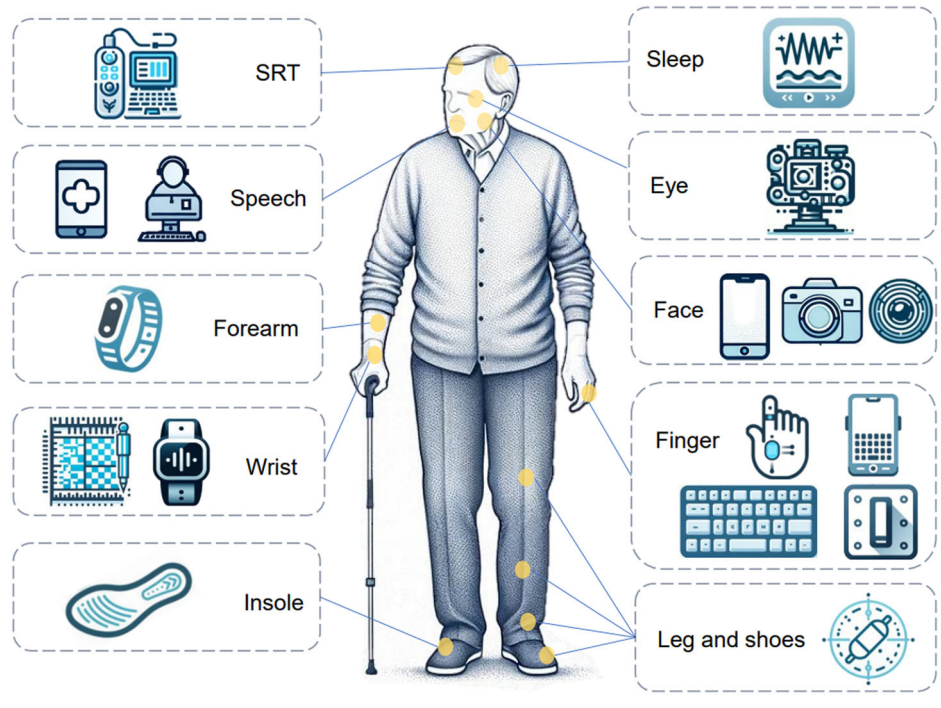
Speech disturbances

Hypokinetic dysarthria occurs in 90% of PD patients, characterized by diminished movement and control of speech-related muscles. It manifests as monotone and reduced loudness, unclear pronunciation and disrupted speech speed and rhythm²⁹. The assessment of language function typically relies on item #18 (language) of Part III of the UPDRS scale⁵. Hypokinetic dysarthria may manifest in the early stages of PD, with acoustic measurements proving sensitive in detecting initial motor symptoms and helpful in quantifying treatment response³⁰. An early voice-testing device, the AHTD developed by Intel, yielded promising results in assessing symptoms that are challenging to detect, supporting UPDRS follow-up in severely affected patients³¹. Almeida et al. developed a PD classification application designed

Table 1 | The differences of digital biomarkers over tissue biomarkers

	Digital biomarkers	Tissue biomarkers
Sample size	Large sample size enhances the robustness of clinical research.	Small sample size, yet strong data availability.
Data collection	Continuity, non-invasiveness	Discontinuity, invasiveness
Location of data detection	Remote measurement, can be conducted at various locations such as homes and workplaces etc.	Hospitals and clinics equipped with professional devices.
Data utility	Used for early prediction, diagnosis, clinical medication guidance and continuously efficacy monitoring of PD.	Diagnosis and alternatively medication monitoring for individuals of PD.
Limitations	Standardization of device data, complexities in data analysis, patient compliance, lack of clinical validation and the limitations in application to remote areas with limited internet coverage.	Differences in detection technology and data processing methods, measurements can only be conducted intermittently in medical environment and poor patient compliance.

Fig. 1 | Devices for measuring digital biomarkers in various body parts of Parkinson's disease patients. This figure illustrates devices used for measuring PD-related digital biomarkers, including wearable devices positioned on different body parts (such as wrist-worn accelerometers, smartwatches, accelerometers worn on the forearm and shoes, smart insoles, etc.), and instruments for measuring motor and non-motor symptoms in various body parts (such as finger-tapping keyboards, cameras for facial monitoring, smartphones, etc.). SRT, serial reaction time.



for installation on smartphones, intended for use by physicians³². Researchers tested voice recordings from speakers of different languages and found high prediction accuracies, mostly between 70% and 90%. However, accuracy dropped when testing across languages due to language-specific models³³. In a longitudinal study, objective acoustic measurements documented progressive deterioration in voice and speech patterns as the disease advanced³⁴. However, voice detection, while effective, faces limitations such as small sample sizes, sample imbalances, inconsistent distances between the mouth and microphone and varied measurement metrics across datasets.

Facial bradykinesia

Facial bradykinesia, characterized by a decrease and deceleration in facial movements, emerges as one of the predominant motor symptoms in PD, leading to challenges in recognizing a PD patient's emotion by his facial movements. PD patients produce more "fake smiles", leading to the impression of apathy and withdrawal, known as "masked" syndrome³⁵, which can exacerbate social withdrawal in PD patients. Factors such as biology, the severity of the disease, cognitive dysfunction and the effects of dopaminergic medication are believed to modulate the capacity for facial movements.

Facial expressions serve as a critical means of conveying emotional states, encapsulating both the "expression" and "perception" of emotions. Prior to the introduction of the Facial Action Coding System (FACS) by Ekman and Friesen in 1977, research into facial behaviors predominantly depended on subjective observations. The FACS framework identifies discrete action units (AUs) within facial expressions, enabling each expression to be described through a specific combination of these AUs. Contemporary approaches to automatic facial motor recognition strive to provide objective characterizations of expressions linked to emotional states. These approaches are generally divided into static and dynamic systems. Static systems often employ "facial mimicry", where subjects are asked to emulate fundamental expressions, highlighting notable disparities in facial movements between PD patients and healthy controls³⁶. On the other hand, dynamic systems, capable of automatically detecting facial AUs, offer enhanced sensitivity in identifying nuanced changes in motor. In recent developments, Jin et al. have pioneered a method for diagnosing PD using facial video recognition technology coupled with machine learning algorithms. This approach has achieved a diagnostic accuracy exceeding 90% for PD, yet

it lacks the capability to differentiate atypical syndromes³⁷. Nonetheless, the broader clinical adoption of this model remains constrained by ethical considerations.

Ocular motility dysfunction

The intricate alterations in cerebral function render eye-tracking an indispensable instrument for delving into progression mechanisms of PD. Approximately 75% of PD patients experience eye movement difficulties³⁸, with PD-induced neuropathological alterations leading to prolonged fixation durations, delayed initiation of saccades and reductions in saccade amplitude and velocity³⁹. Saccades are commonly utilized to assess eye movement abnormalities in PD patients. Research indicates that PD patients display shorter maximum fixation times and elevated saccadic rates, suggesting a heightened susceptibility to distraction⁴⁰. Key digital biomarkers for saccades encompass frequency, count, duration and amplitude. Furthermore, visual cues provided by smart glasses can enhance gait outcomes in PD patients⁴¹. Multi-step saccades (MSS), particularly in vertical saccades, complicate the clinical distinction between Progressive supranuclear palsy (PSP) and PD during the early stages⁴². Herwig et al. have effectively differentiated these diseases using head-mounted displays and computing an index by multiplying the ratios of vertical to horizontal gain and velocity⁴³. The impaired capacity of PD patients to suppress reflexive behaviors necessitates the assessment of anti-saccade movements as an additional diagnostic criterion. Relevant metrics include anti-saccade latency, delay and error rate. Despite variability in findings across studies, a meta-analysis by Waldthaler et al. corroborated significant increases in anti-saccade latency and error rates in PD patients⁴⁴. Villers-Sidani et al. distinguished PD from healthy controls with high precision through a comprehensive analysis of fixation, saccade and anti-saccade tasks using tablets and software⁴⁵.

Abnormal pupil size and blinking abnormalities

Autonomic dysfunction may represent an early sign of PD, with recent research indicating modifications in blink and pupil patterns in the prodromal phase, unlike changes in other saccadic movements⁴⁶. The pupillary light reflex (PLR) serves as a crucial marker of pupillary functionality, with the parasympathetic nervous system being more significantly impacted than the sympathetic nervous system in PD individuals⁴⁷. This dysregulation

Table 2 | The application of digital biomarkers in Parkinson’s Disease

Functionality	Classification	Digital biomarkers	Detection devices	Participants	Effects and Sensitivity/Specificity	References
Serial reaction time (SRT)	SRT-ST/DT	Response time after an event occurs	Serial Response Box (SRBox, model 200a, Psychology Software Tools Inc., Pittsburgh, PA, USA); E-Prime software (Psychology Software Tools Inc., Pittsburgh, PA, USA)	39 PD, 39 HC	PD patients responded more slowly overall than HC ($P = 0.001$).	18
		Implicit and explicit sequence learning	A personal computer with a 33-cm color monitor connected to a four-button box	13 PD, 15 HC	PD patients’ performance on implicit and explicit sequence learning tasks are worse than HC.	19
	EEG headband	Total sleep time, sleep efficiency, wakefulness after sleep onset, light sleep duration, deep sleep duration and REM sleep duration	Dreem Headband, Version 2	10 PD	Compared to polysomnography, the specificity was higher than sensitivity when analyzed epoch by epoch, demonstrating sufficient accuracy (0.55-0.82) in identifying sleep stages.	24
Sleep	Nighttime respiratory signals	Respiratory signals	Chest or abdominal breathing band, radio equipment	757 PD, 6914 HC	Sensitivity 80.22% and specificity 78.62% for breathing belt, and sensitivity 86.23% and specificity 82.83% for wireless data.	25
	Wearable device	Heartbeat intervals (RR intervals) were used to classify sleep stages	ASUS Vivo Watch BP	20 PD (10 with and 10 without clonazepam), 18 HC	A statistically significant difference in the percentage of abnormal REM between the control group and the PD group ($P = 0.004$).	27
	Smart bed	Respiratory rate, heart rate and posture	Bed sensor and an application for Android phones, iPhones and Android/iOS tablets	-	-	28
Speech		Frequency (tremor), Amplitude (flutter), HNR, NHR, RPDE, DFA, PPE	Head-mounted microphone and Intel AHTD remote monitoring system	52 early PD	The potential of predicting average PD symptom progression, and establishing a mapping between dysphonia measures and UPDRS.	31
		Phonation (P), Speech (S), Unvoiced part (U) and Voiced part (V)	Acoustic cardioid (AKG Perception 220, frequency range 20–20,000 Hz) and a smartphone (an internal microphone of Samsung Galaxy Note 3)	64 PD, 35 HC	The best performance for the AC channel achieved an accuracy of 94.55%, AUC 0.87. When using the SP channel, we have achieved an accuracy of 92.94%, AUC 0.92.	32
		Voice, articulation, fluency, prosody, NHR, jitter, shimmer, mean fundamental frequency	An audio software (Steinberg Wavelab/Steinberg Media Technologies GmbH, Hamburg, Germany) and a headset microphone	80 PD, 60 HC	The pronunciation and language of PD patients get worse as the disease progresses ($P < 0.001$).	34
Facial movement	Static identification	Basic expressions (Neutral, Happy, Sad, Contempt, Surprised, Angry, Disgusted and Fearful)	Web camera, and Microsoft Azure Face (MicrosoftCorp., Washington, USA)	96 HC, 97 PD	Expressionless was obviously more frequent and happiness was less frequent in PD than in control ($P < 0.05$).	36
	Dynamic identification	Facial muscle expressions and tremor intensity	Canon 700D camera	33 PD, 31 HC	The accuracy of classifying facial expressions using widely used machine learning algorithms is above 90%.	37
Ocular motility	Saccade	Saccades (frequency, number, duration, amplitude) and fixations (duration)	Mobile infrared eye-tracker (Tobii Pro Glasses 2, Tobii, Inc.)	17 HC, 21 PD without FoG, and 22 PD with FoG	Visual cues produced significant improvements in gait compared to usual walking across all groups.	41
	MSS	Saccadic latency, mean and peak velocity, gain, AS error rate as well as rates of express and anticipated saccades	A head-mounted binocular eye-tracker: EyeBrain T2 (medical device with CE label for clinical use Class IIa, ISO 9001, ISO 13485)	40 PD, 20 HC	The research demonstrates a sensitivity of 68% and a specificity of 90% in effectively distinguishing between patients with PD and HC.	42
	AS	Direction error rate, direction corrected rate, target (arrow) recognition rate, correct direction latency and incorrect direction latency	a 12.9-inch iPad Pro tablet with the ETNA™ software installed	59 PD, 62 HC	It was able to accurately distinguish PD patients from healthy controls with a sensitivity of 0.93 and specificity of 0.86.	45

Table 2 (continued) | The application of digital biomarkers in Parkinson’s Disease

Functionality	Classification	Digital biomarkers	Detection devices	Participants	Effects and Sensitivity/Specificity	References
Pupil size	White light stimulation	Latency, duration, velocity, and amplitude of pupil response to short/long flash stimulation	A screen-based eye tracker (Tobii Pro Spectrum)	36 PD, 35 HC	The eye tracker can robustly evaluate the differences in pupil size between patients with PD and HC ($P < 0.001$).	76
	Red-blue light stimulation	Measurement of PIRP	An infrared sensitive XIMEA MQ013RG-E2 machine vision camera and XIMEA CamTool software	24 PD, 11 HC	The measured PIRP indicating different responsiveness to the wavelengths between PD and HC and demonstrating a highly significant difference.	77
Blinking		Saccades, AS, pupil and blinking behavior	An infrared video-based eye tracker (EyeLink 1000 Plus, SR Research Ltd, Ottawa, ON, Canada)	121 PD (45 Normal/45 MCI/20 Dementia/11 Other), 106 HC	The classifier reached a sensitivity of 83% and a specificity of 78%.	51
Finger tapping	Button-style sensor	Typing frequency at different typing speeds	A board mounted with a metal switch	51 PD (14 TP, 11 FP, 12 AP, 14 UC), 36 HC	PD patients tapped at a significantly slower rate than controls when asked to tap at their fastest rate ($P < 0.01$).	78
	Accelerometer	Velocity, Amplitude, Standard deviation of intervals	3-axis accelerometers, touch sensors, AD converter and personal computer	16 PD, 27 HC	The system shown that the acceleration and output of the touch sensor could be measured and the features could be extracted.	5
	Video recording	Amplitude, frequency and velocity	Smartphone (iPhone X*)	26 early PD, 64 HC, 9 idiopathic anosmia	Combining both velocity mean and the CV of frequency meant that the specificity 86.67% for the same sensitivity (AUC 0.83; 95% CI 0.72 to 0.95).	53
	Simple keyboard typing	HT, FT and pressure sequences	Keyboard equipped with the iPrognosis application	27 PD, 84 HC	It has shown a classification performance with an AUC of 0.79 and sensitivity/specificity of 0.74/0.78 in PD and HC.	79
Wrist movement	Text excerpt typing	HT, FT and NP	Android smartphone (LG Nexus 5X with a screen of 5.2 inches in diagonal and a resolution of 1080x1920 pixels, running native Android 7.0)	18 early PD, 15 HC	This method achieved an AUC = 0.92 and 0.82/0.81 sensitivity/specificity.	56
	Wrist FE	Wrist RUD and FE	Sensor (PS25454 EPIC, Plessey semiconductors, UK)	29 PD, 29 HC	The wrist movement variability is greater for the HC than for the PD and it can help to discriminate between both groups ($P < 0.05$).	80
	Motion controller	Volume, velocity, acceleration, and frequency of hand positions (25 joints, including all fingers/thumb and wrist)	A Leap Motion Controller (LMC), (Leap Motion, Inc., San Francisco, CA, USA)	55 PD	The analysis showed that the volumes, acceleration, and velocities had significant correlations with clinical MDS-UPDRS scores.	81
Forearm motor	Accelerometer	Acceleration and angular velocity	A triaxis accelerometer sensor (LS3DSH, STMicroelectronics, Switzerland) and triaxis gyroscope sensor (L3GD20, STMicroelectronics, Switzerland)	85 PD	The method has an accuracy of 85.5%.	63
	Tremor	Continuous inertial sensor data	Apple Watch and its application MM4PD	343 PD, 171 elderly HC	MM4PD measurements correlated to clinical evaluations of tremor severity ($P < 0.001$).	64
Forearm motor	Continuous local discharge monitoring	AC voltage signal generated by friction	Ecoflex™ and PEDOT:PSS	-	The sensor platforms have demonstrated a good sensitivity to PD symptoms like bradykinesia and tremor based on the UPDRS.	82
Gait and balance	Waist	Freeze index for each direction (X, Y, Z) as well as the vector products of XY, XZ and YZ for all walking and turning tasks	A triaxial accelerometer (DynaPort MiniMod, McRoberts)	18 PD with FoG and 5 PD without FoG	Combining full rapid turns and walking with short steps rapidly tasks rendered a sensitivity of 75% and specificity of 76%.	83
	Thigh	Vertical acceleration	3D-acceleration and an Intel XScale family processor	10 PD	The theoretical maximum performance of this algorithm is 88.6% sensitivity and 92.8% specificity.	84
	Shank	Vertical linear acceleration and pitch angular velocity	Vertical linear acceleration	11 PD		85

Table 2 (continued) | The application of digital biomarkers in Parkinson’s Disease

Functionality	Classification	Digital biomarkers	Detection devices	Participants	Effects and Sensitivity/Specificity	References
Ankle		The angular velocity of the z-axis and the xy-plane	Three-axial gyroscopes and accelerometers (Rehacom®, Hasomed, Magdeburg, Germany)	23 PD, 13 HC	Individual calibration of the freeze threshold improved accuracy and sensitivity of the device to 89% for detection of FoG with 10% false positives.	86
			The sensor-based algorithm exhibited a specificity of 98%, a sensitivity of 85%, and an accuracy of 0.96 for the detection of dyskinesias.			
Footwear		Stride length, stride duration, gait speed, cadence, stance phase duration, swing phase duration, HS angle, TO angle, gait variability and foot clearance	Two sensor units (Shimmer Sensing, Dublin, Ireland), including a tri-axial accelerometer (Freescall Semiconductors MMA7361) and a tri-axial gyroscope (InvenSense 500 series)	190 PD, 101 HC	Gait parameters were sensitive to changes by mirroring the progressive nature of PD.	87
Insole		Pressure values of the sensors, acceleration and angular velocity values for each sensor in the xyz coordinates	Motion ReGo instrumented insoles (specifically, Model Insole 3, Munich, Germany)	19 PD	The AUC values of the classifier trained using data features from the insole instrument range from 0.85 to 0.92, with an accuracy of 88–91%.	88
FoG		FoG, temporal markers for the onset and cessation of halting and turning movements during ambulation	3-axis accelerometer and gyroscope	28 PD	DeepFoG achieves 88% and 90% sensitivity/specificity.	69
Accelerometer	UK Biobank		Activity AX3 wrist-worn triaxial accelerometer	43,730 HC, 273 PD, 196 Prodromal PD	The machine learning model exhibits excellent testing performance in diagnosing both prodromal PD and clinical PD.	66

ST single tasks, DT dual tasks, PD Parkinson’s disease, HC health control, EEG Electroencephalogram, REM rapid eye movement, HNR Harmonic-to-Noise Ratio, RPDE recurrence period density entropy, DFA detrended fluctuation analysis, PPE pitch period entropy, AHTD the Association for High Technology Distribution, UPDRS Unified-Parkinson Disease Rating Scale, AC professional microphone, SP smartphone, AUC area under curve, FoG freezing of gait, MSS Multiple step saccade, AS Anti-saccade, PIPR post-illumination pupillary response, MCI mild cognitive impairment, TP tremor predominant FP freezing predominant, AR akimetic/rigid, UC unclassified group, HT hold time, NP normalized pressure, FE flexion and extension, RUD radial and ulnar deviation, HS heel strike, TO toe-off.

between the parasympathetic and sympathetic nervous systems can precipitate atypical pupillary responses in the disease’s early stage, primarily assessed through direct measurement of patient pupil responses to light. It has been observed that PD patients demonstrate reduced speed and efficacy in pupil dynamics. Joyce et al. identified that the postillumination pupillary response (PIPR) can serve as an early PD assessment tool⁴⁸. Tabashum et al. developed a clinician-oriented system for the automated monitoring of pupil diameter changes to evaluate PIPR, thereby establishing PIPR as a dependable digital biomarker for early-stage PD diagnosis⁴⁹. Furthermore, PD patients exhibit a decreased blink rate and excitability, a condition that is possibly related to the depletion of dopaminergic neurons, yet frequently ameliorates with the administration of levodopa or amantadine treatment⁵⁰. Brien et al. utilized video-based eye tracking in a multifaceted task to concurrently assess saccades, pupil and blink characteristics, integrating machine learning models to estimate PD diagnostic confidence scores⁵¹.

Dysfunction in upper extremity movement

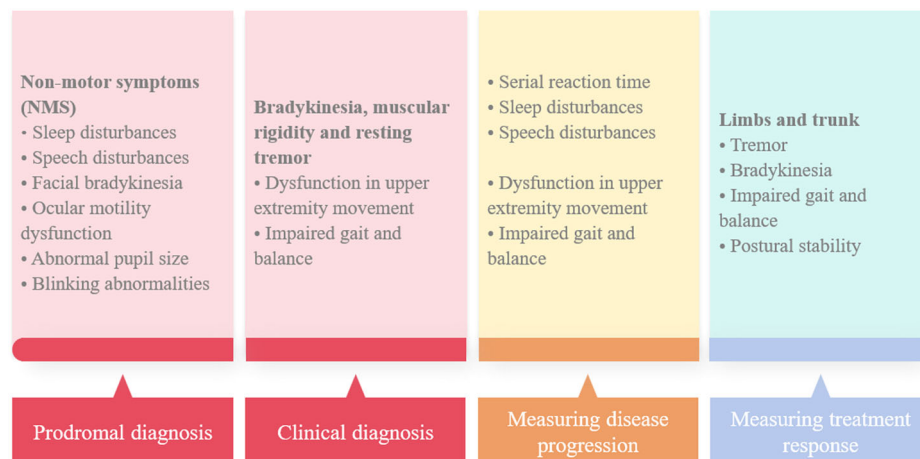
In the majority of instances, the motor symptoms of PD first manifest in the upper limb, significantly impacting the patient’s ability to control and coordinate these movements crucial for daily activities. The UPDRS Part III (Motor Examination), particularly items #23 (finger tapping), #24 (hand movements) and #25 (rapid alternating hand movements), serves as the gold standard in clinical settings for evaluating the severity of upper limb motor symptoms⁵. Consequently, digital biomarkers assessing upper limb movement through the analysis of fingers, wrists and forearms have gained prominence.

Degeneration of the basal ganglia in PD leads to a higher frequency of tapping with shorter intervals and greater variability⁵². While traditional observation may not capture all nuances, digital methods provide quantitative data that support ongoing research. Multiple methodologies exist for measuring finger tapping (Table 2). Finger tapping can be measured using keyboard inputs or touchscreen interactions, while video analysis has emerged as a highly accurate method for evaluating PD. Williams and Simonet et al. used contact-free smartphone videos to quantify Parkinsonism-related measures in finger tapping, demonstrating high diagnostic accuracy and potential for monitoring medication efficacy^{53,54}. Alternating Finger Tapping (AFT) represents another variation, with mPower, a study leveraging Apple’s ResearchKit library, investigating AFT by having participants tap two fixed points on a screen alternately with one hand’s fingers⁵⁵. Iakovakis et al. diagnosed early PD patients, using dynamic keystroke information from natural typing on touchscreen smartphones, based on brief text excerpts by participants⁵⁶. Nevertheless, the influence of confounding variables, such as patients’ sleep quality and their proficiency in computer usage, necessitates further exploration into machine learning algorithms and mobile applications that can enable more effective clinical applications.

Rigidity is a predominant symptom of PD, affecting approximately 89% of patients⁵⁷. Both neural reflexes and intrinsic mechanisms are believed to contribute to rigidity in PD⁵⁸, with the wrist being particularly affected. Activities essential for daily life, such as eating, drinking and reading, involve movements ranging from 35 degrees of flexion to 10 degrees of extension. Wrist flexion and extension (FE) are crucial for assessing motor signs in PD patients, with radioulnar deviation (RUD), pronation and supination being secondary movements. Sensors detecting upper limb movement placed on the wrist and forearm (Table 2) exhibit good sensitivity to PD symptoms like bradykinesia and tremor.

The use of wrist-worn accelerometers stands out, with early detection through wearable technology becoming a hot topic in research. From the social cognition perspective, wrist is the optimal site for wearing and interacting with electronic devices⁵⁹, minimizing loss to follow-up rates. Numerous studies have aimed to detect specific PD features such as tremor, freezing of gait and bradykinesia (see impaired gait and balance)⁶⁰. To assess patients’ motor function at home, the Verily team created a PD motor exam using the Verily Study Watch⁶¹, launching the Parkinson’s Disease Project (PPP) clinical trial⁶². Jeon et al., using a watch-type wearable device analyzed

Fig. 2 | The utilization of digital biomarkers throughout various stages of Parkinson's Disease: an idealized representation not yet fully supported by current evidence. This figure shows the different roles of various digital biomarkers in the prodromal diagnosis, clinical diagnosis, disease progression measuring, and therapeutic efficacy measuring of PD in an ideal scenario. Due to their current clinical application, this is not yet fully supported by current empirical evidence.



and proposed an alternative method for developing an automatic scoring system for PD tremor using a machine learning approach⁶³. The tests' reliability were found to be consistent with MDS-UPDRS III and sensitive to drug effects. Powers et al. have developed a sophisticated dynamic monitoring system based on Apple Watch, known as the Movement Monitor for Parkinson's Disease (MM4PD), which is adept at accurately capturing and monitoring the patterns of resting tremors⁶⁴. Nonetheless, the limitations of this research include substantial costs, potential errors arising from reliance on single observational points and biases associated with a limited sample size of participants.

Impaired gait and balance

PD is characterized by notable gait and balance impairments, manifesting as reduced stride lengths, diminished gait velocities, and an increased susceptibility to falls compared to healthy individuals. Nonetheless, detecting gait alterations in PD's initial stages proves challenging due to their subtlety. Moreover, assessments of motor fluctuations in early-stage PD often rely on patient-reported durations of on-off periods, subject to perceptual and recall biases. Wearable devices enable continuous, unobtrusive monitoring of movement patterns, mitigating the impact of subjective gait variability among participants. Among the applications of wearable accelerometers, the study of freezing of gait (FOG) is prominent, though their use extends beyond this domain.

Wearable devices for gait detection, positioned on the waist, thighs, calves, ankles or integrated into shoes (Table 2). Wearable devices can be strategically positioned on various body segments. Mancini et al. employed accelerometers affixed to the posterior trunk, proximal to the body's center of mass, and on the anterior lower legs for conducting the Instrumented Timed Up and Go test (ITUG). This configuration facilitated enhanced discrimination of gait disturbances among PD subjects, distinguishing those with and without FOG with greater precision⁶⁵. Given the synchronized movement of arms and legs during ambulation, wrist-worn accelerometers also serve to gather gait-related digital biomarkers in PD. Schalkamp et al. utilized UK Biobank accelerometer data to underscore its potential as a cost-effective screening tool for PD risk identification⁶⁶. Fitbit devices also offer sufficient accuracy and precision for estimating health metrics in PD patients, including step count, calories burned, active minutes and sleep stages⁶⁷. Additionally, Global Kinetics Corporation has developed a wearable device called the Personal KinetiGraph (PKG), which not only detects unperceived bradykinesia or dyskinesia in patients but also provides medication dosage reminders to the patients⁶⁸. PD patients experiencing FOG often avoid multitasking with their arms while walking, focusing instead on their stride and forthcoming steps. However, inducing FOG episodes during clinical evaluations may be challenging, necessitating continuous assessment through wrist-worn accelerometers⁶⁹. Bikias et al. demonstrated the high specificity of the DeepFoG method, which employs training based on

deep learning models, in reducing the risk of falls, which highlights its ecological validity⁶⁹. Wearable interventions for assisting PD patients with FOG aim to detect episodes in real-time and employ sensor data, such as from accelerometers, to provide immediate rhythmic cues that facilitate resumption of walking⁵⁹.

Adjunctive therapy

Digital biomarkers offer a practical approach to the remote monitoring and continuous intervention of patients, significantly enhancing the personalization and accessibility of patient consultations and improving therapeutic outcomes. The pioneering study by Page et al. was among the initial endeavors to evaluate the effectiveness of PD therapies using digital instruments, incorporating smartphones in clinical research to pave the way for their future application in clinical trials. However, the study revealed only a moderate correlation due to reduced compliance with smartphones during the experiments and low data acquisition rates⁷⁰. Di Lazzaro et al. employed wearable sensors to demonstrate that dopaminergic replacement therapy significantly improves motor functions beyond tremor control, and notably increases the sensitivity in detecting early subclinical abnormalities in gait and lower limb movements⁷¹. Powers et al. integrated the MM4PD software on the Apple Watch and precisely captured motor fluctuations and medication responses in 94% of evaluated subjects, delineating more detailed on-off patterns throughout the day, and establishing this integration as a vital tool for clinical decision-making in medication titration⁶⁴. The clinical efficacy of deep brain stimulation (DBS) for the treatment of PD has been well established. Standard practices for setting DBS parameters often rely on subjective assessments. Pulliam et al. developed an algorithm that leverages motion sensors attached to fingers to collect movement data, which is subsequently utilized to fine-tune DBS parameters, potentially diminishing the severity of PD symptoms by an additional 10%⁷². However, the implementation of such technologies necessitates meticulous consideration and stringent evaluation.

Future of digital biomarkers

The digital transformation of healthcare in recent years has significantly advanced digital technologies, facilitating the transition of digital biomarkers from research contexts to clinical settings. Nevertheless, incorporating digital technology into healthcare comes with its set of challenges. Foremost among these are regulatory concerns regarding privacy, as well as the necessity for thorough testing of feasibility and reliability. In clinical practice, the limited sensitivity and specificity of certain fields necessitate validation of its clinical efficacy. Furthermore, the proprietary nature of algorithms, with only a select few vendors offering textual descriptions of their coding, device compatibility issues between different manufacturers

and models raise doubts about the accuracy of the mathematical algorithms used for data processing. Although algorithmic accuracy can improve with extensive data collection, variations in measurement conditions can impact data accuracy. Data collation and analysis require the extraction of meaningful signals from background noise, and there is a lack of standardized data verification protocols. Patient adherence to long-term device usage is often poor (e.g. mPower), and remote studies commonly suffer from low participant retention rates, exacerbated by the lack of mature technology for ensuring patient privacy^{73–75}. To address these issues, which present substantial barriers to the broad clinical adoption of these technologies over an extended period, is an urgent priority for future research.

Additionally, the development of algorithms that integrate multiple digital biomarkers for PD is necessary to enhance diagnostic accuracy. To deliver maximal patient benefit, a secure and effective digital biomarker ecosystem requires algorithmic transparency and interoperable components with open interfaces, facilitating the rapid development of new multi-component systems and high-integrity measurement systems⁹. Extensive clinical trials are also imperative to validate their accuracy. The swift advancements in digital health technologies enable the continuous collection of health data in natural settings, and the enhancement and transparency of algorithms further catalyze the clinical integration of digital biomarkers. Looking ahead, it is anticipated that the expansion of digital biomarker databases and algorithmic innovation will bolster the precise diagnosis of PD, monitor treatment effectiveness and inform therapeutic decisions.

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Author contributions

Y.S. wrote the initial draft. C.S. developed the concept for the article. Z.W., Y.L. and C.H. contributed important ideas and participated in reviewing and editing of the text.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Chang-he Shi.

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